Optimization of Deep Neural Network Model for Embedded Applications: A Case Study of Retinal Vessels Segmentation

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***Abstract*— Medical science is an area where deep neural networks (DNNs) have been successfully applied for disease diagnosis based on image analysis. The accurate diagnosis of different eye diseases requires retinal vessels segmentation in retinal fundus image. The focus of this work is the improvement of innovative, lightweight deep learning (DL) models that can be used on embedded systems (ESs) for eye diseases diagnosis in the early stage at the point of care in clinics and general hospitals. This work aims to optimize the encoder-decoder-based architecture to show the possibility of porting a DL model onto ESs but without compromising on segmentation performance. The performance was evaluated based on open-source retinal image databases: DRIVE, and CHASE_DB1. The proposed model achieved encouraging evaluation metrics in addition to attributing a significantly lower number of parameters. Moreover, the proposed model outperformed the previous related works while its computational complexity is much lower.**

Keywords*— *Embedded System, Diabetic Retinopathy, Glaucoma, Image Segmentation, Lightweight Deep Learning

I. INTRODUCTION

Researchers have developed segmentation algorithms to increase the accuracy of segmentation for various organs since Image segmentation is regarded as the first step toward developing effective strategies for image-processing-based disease diagnoses [1]. Diagnosis of the structure of the retinal vascular tree and the optic cup/disc is critical for identifying various eye diseases such as diabetic retinopathy (DR), glaucoma, and age-related macular degeneration (AMD).

Diabetes-related retinal vasculopathy (DR) is described as a dysfunction in the retinal vasculature induced by long-term diabetes. DR causes blindness before the age of 50, with essentially no symptoms in the early stages [3]. Glaucoma is a long-term condition that damages the optic nerves. Glaucoma can harm blood vessels and optic nerves if it is not recognized and treated promptly [4]. One of the most prevalent causes of visual loss is AMD. It affects persons aged 65 and up and the pathogenesis remains unclear [5].

Traditionally, ophthalmologists manually study and analyze retinal fundus images, which is hard and time-consuming for experts [2]. Additionally, manual analysis cannot be scaled for population-level screening. The remarkable advances in the field of image processing, computer vision, ML, DL, and big data have led to the possibility of developing and using computer-aided diagnosis (CAD) tools for diagnosing different diseases. Consequently, an advanced framework based on automated or semi-automated CAD tools plays an important role in bio-medical image processing and exceeds the performance of traditional segmentation methods in terms of various evaluation metrics [6]. Advanced CAD tools have the potential to alleviate the strain on public healthcare resources dramatically.

The most frequently studied and trended technology among academicians and researchers these days are artificial intelligence (AI), machine and deep learning. These emerging technologies have sparked the fourth industrial revolution in human history [7]. Thanks to the huge development in the fields of processing power and the cheap availability of graphical processing units (GPUs) in recent years. Equally, the availability of big data both in terms of the electronic health record and the well-maintained different disease-related medical image databases has also played an unavoidable role.

Deep Neural Networks (DNNs) [8] are highly parameterized, which enables the development of highly expert and reliable models for accurate predictions in a wide range of real-life applications. Moreover, a variety of neurons with distinct functions have been suggested for the DNN models. One such most commonly applied model created by combining DL and image processing technologies is the convolutional neural network (CNN). CNNs have attained marvelous success in processing 2D and 3D images [9] in numerous real-life applications spanning from healthcare to remote sensing.

Despite their high accuracy, DNNs are more computational and memory-demanding due to their hidden layers increasing the number of parameters. The combination of AI with embedded systems (ESs) marks the possibility of implementing operations that require real-time processing. Due to a lack of resources, ESs are still not effective enough to support the training of the DNN models. Simple integration would therefore result in ineffective solutions.[10].

This work aims to optimize encoder-decoder-based architecture to show the possibility of porting a DL model onto any general embedded platform. This is based on our most recent exploration of the lightweight DNN model for retinal vessel segmentation. So, the same case study of retinal

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vessel segmentation is considered but the focus is on exploring different possibilities for making the DL model realizable on the embedded platform. In simple words, the exploration and optimization of a lightweight DL model are considered, which is studied and evaluated for the case study of retinal vessel segmentation. The work's contributions are as follows.

- Optimized a DL model with the aim of making it portable to embedded platforms.
- Reduced the memory and computation overhead of the DL model.
- Reduced parameters significantly without compromising on the performance metrics of the DL model.
- Trained and tested the optimized model using two publicly available retinal image databases.
- Performed comparison with the best model from the state of the art using standard evaluation metrics.

The following is the organization of the remaining paper. Section II views the relevant works. Section III describes the enhanced method in detail. Section IV presents the findings and discussion. Finally, section V provides the conclusion of the work.

II. Related Work

With flourishing high-performance computing (HPC), DL, and image processing, numerous architectures for semantic segmentation of biological pictures, such as retinal fundus images, have been developed by researchers. However, those methods generally require significantly higher computation and memory resources, which are generally not available on the general-purpose computers (GPC) used at hospitals. The accurate detection of retinal structures in fundus images needs to develop lightweight DL models that can be implemented on GPC and ESs. Thus, it can be used for early-stage diagnosis of eye diseases at the point of care in hospitals [11].

The authors in [12] developed DNN model called the (CARE) system, which tested fundus images for detecting the most common retinal abnormalities. The performance of CARE was extensively validated with 21867 images and tested using 18136 images. For the validation and testing, they achieved an area under the receiver operating characteristic curve (AUC) of 0.955 (SD 0.046) and 0.965 (0.035) respectively.

The ColonSegNet was developed by the authors in [13], which they used for cancer diagnosis. The ColonSegNet architecture is a lightweight and can be used for real-time implementation. The authors utilized it for real-time segmentation of colonoscopic images, which can detect and locate polyps more quickly. It is a lightweight model and suitable for embedded AI applications.

The authors in [2], suggested the (RCED-Net) DL model for retinal vessel segmentation. Between the encoder and decoder architecture, they applied a residual connection. This model enhanced memory efficiency and reduced the training and testing time. It also decreased the number of tunable hyperparameters.

The authors in [14] suggested LadderNet, which is considered a chain of numerous U-Nets. This approach utilizes a residual block with shared weights and several pairs of encoder-decoder branches, which drastically decreases the number of parameters.

[15], the authors proposed two architectures, both are an extension of U-Net architecture, (RU-Net) and (R2U-Net). The performance of these two models was evaluated on three different datasets: retina blood vessels, lung lesions, and skin cancer segmentation. These models showed impressive results in improving the performance in segmentation tasks while maintaining the same number of parameters without increasing them.

In our recent work in [11], we assessed the performance of ColonSegNet [13] for retinal vessel segmentation using standard retinal image databases including STARE, DRIVE, and CHASE_DB1. It achieved considerably enhanced performance while its computational complexity is significantly lower compared with previous related works. In the current work, we are aiming to reduce the computation complexity of the model in [11] further. The focus is on the development of a lightweight model that can be used in handheld embedded platforms at general hospitals and private clinics for eye disease diagnosis.

III. Experiments

A. Basic Model Architecture and Details

Based on ColonSegNet [13], we recently [11] evolved an accurate and lightweight model for retinal vascular segmentation. It is an Encoder-Decoder architecture developed for real-time polyp segmentation. The block diagram in Fig. 1. presents the architecture of ColonSigNet [13]. This model offers an attractive feature, particularly for applications using computationally restricted devices, such as the early diagnosis of various eye diseases in clinics and general hospitals for population screening.

B. Enhanced Model and Details

In the experiments, we tested the effect of changing the number of filters on the numerous standard evaluation metrics of the developed model. We tried different model architectures by changing the number of filters. It is important to mention that the number of filters (also known as convolution kernels) has a deeper impact on both the performance of the model and the number of its parameters. Our aim has been to reduce the number of parameters in order to reduce the computational complexity without compromising on the performance metrics. To obtain that, it is specified that the number of filters is either 32 or 64.

The newly developed DL model based on varying the convolutional kernels is fully trained from scratch.

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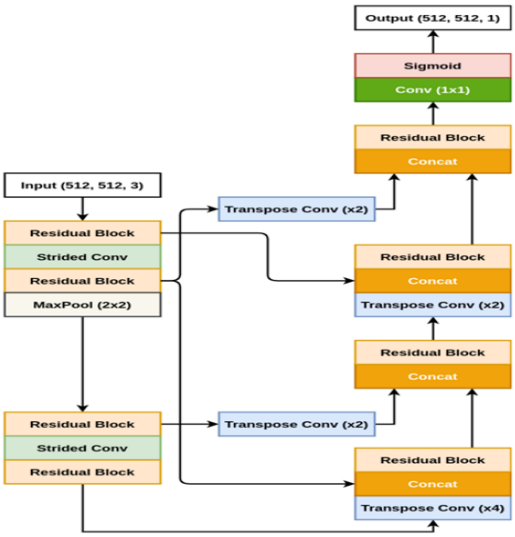


Fig. 1. ColonSegNet block diagram [13]

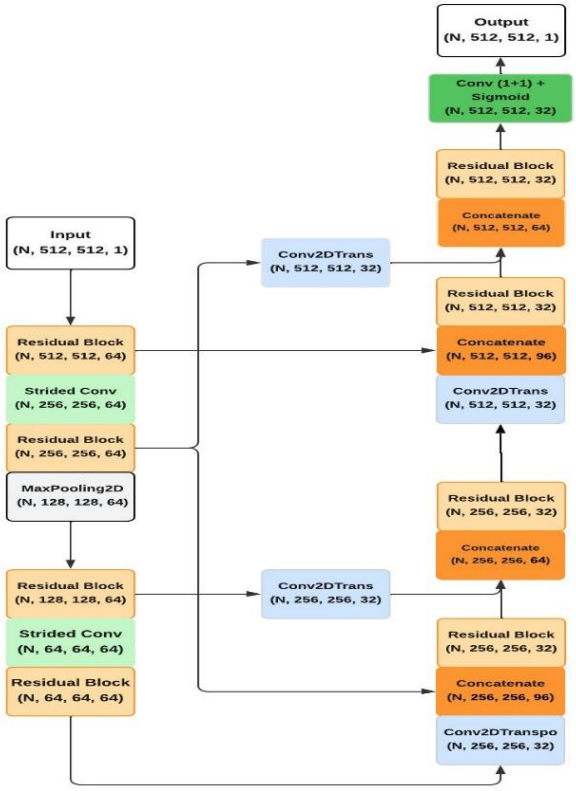


Fig. 2. Number of filters of RetSegNet

We compared the performance of the proposed model to prior best models in the literature using standard evaluation metrics. The newly developed model is termed RetSegNet, which is depicted in Fig. 2.

In this experiment, we tried decreasing the number of filters. Starting with 64 filters in the Encoder parts and then decreasing to 32 in the Decoder parts. In the first Encoder, we kept the number of filters to 64, while in the second Encoder, we reduced the number of filters from 256 to 64. In addition, 64 filters are assigned to Max-pooling. In the first Decoder, we reduced the number of filters from 128 to32. All residual blocks and transpose convolution contain 32 filters. The second decoder, all blocks decreased to 32 filters.

IV. Results And Discussion

This work was inspired by our recent work [11], It achieved considerably enhanced performance although its computational complexity is substantially lower as compared to the existing state-of-the-art. This proposed work decreased the number of parameters from 5.00M to 607,777, compared with [11]. In this section, we studied the effect of changing the number of filters on the model performance and the number of its parameters. The results compared with existing studies models, which are included in Table I, and Table II. (The findings for the preceding models were obtained from their respective papers). The best and second-best results are indicated in green and blue column fields.

A. The Quantitative Performance Comparison

The developed model's performance was assessed using standard evaluation metrics such as accuracy, specificity, sensitivity, AUC, and MCC. This study makes use of two open-source retinal imaging datasets: DRIVE and CHASE DB1. As demonstrated in Table I, the developed model produced promising results for the DRIVE database when compared to other studies. RetSegNet has the highest sensitivity, accuracy, AUC, and MCC, whereas ColonSegNet [11] is the second best. R2U-Net [15] has the highest specificity, followed by Ladder-Net [14] in second place. For the CHASE_DB1 database, Table II reveals that in terms of sensitivity and AUC, RetSegNet, and ColonSegNet [11] have come in first and second place. The accuracy of ColonSegNet [11] is the best followed by RetSegNet comes in second place. R2U-Net [15] has the highest specificity followed by Ladder-Net [14].

B. The Qualitative Performance Comparison

In this section, we qualitatively assessed the experiment's performance based on three different segmented images that were selected for each of the DRIVE and CHASE DB1 datasets. The visual comparison of the developed model with ColonSegNet [11] is shown in Fig. 3 based on the segmented images 7, 11, and 19 from the DRIVE database.

TABLE I. DRIVE Dataset Compariso Results

	Year	Se	Sp	Acc	AUC	MCC
R2U-Net [15]	2018	0.7792	0.9813	0.9556	0.9784	_
Ladder-Net [14]	2018	0.7856	0.9810	0.9561	0.9793	_
RCED-Net [2]	2020	0.8252	0.9787	0.9649	0.9780	_
ColonSegNet [11]	2022	0.8491	0.9774	0.9659	0.9850	0.7960
RetSegNet	2022	0.8507	0.9779	0.9667	0.9858	0.7998

TABLE II. Comparison results on the DRIVE dataset

	Year	Se	Sp	Acc	AUC	MCC
R2U-Net [15]	2018	0.7756	0.9820	0.9634	0.9815	_
Ladder-Net [14]	2018	0.7978	0.9818	0.9656	0.9839	_
RCED-Net [2]	2020	0.8440	0.9810	0.9722	0.9830	_
ColonSegNet [11]	2022	0.8607	0.9806	0.9731	0.9869	0.8014
RetSegNet	2022	0.8729	0.9794	0.9727	0.9872	0.7902

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The ground truth images are shown in the second column. Columns 3 and 4 indicate the outputs of ColonSegNet [11] and RetSegNet. According to our observations of these images, the obtained results from RetSegNet could discover the small vessels better than ColonSegNet [11]. Fig. 4 shows the vessel segmentation analysis for images 2, 5, and 8 from the CHASE DB1 dataset. show the image and the image's ground truth respectively. Columns 3 and 4 show the outcomes of ColonSegNet [11] and RetSegNet. Fig. 4 shows that, as ColonSegNet [11], RetSegNet could detect most of the small details with very little noise.

V. Conclusion

The developments in biomedical image segmentation became possible with unparalleled advancement in the fields of ML/DL, HPC, Big Data, and image processing. These DL-based techniques typically require higher computational and memory resources. Accurate diagnosis of retinal vessels in fundus requires the development of innovative, lightweight DL models that can be used on GPC and embedded platforms. Therefore, different eye diseases can be detected in the early stages at hospitals. In this work, we optimized an Encoder-Decoder architecture for retinal vessel segmentation by reducing computational complexity and parameter numbers while improving performance. The developed model's performance produced a highly convincing outcome. compared with other lightweight and complex methods based on the standard evaluation metrics. This model offers an attractive feature, particularly for applications using computationally restricted devices , such as the early diagnosis of various eye diseases in clinics and general hospitals for population screening.

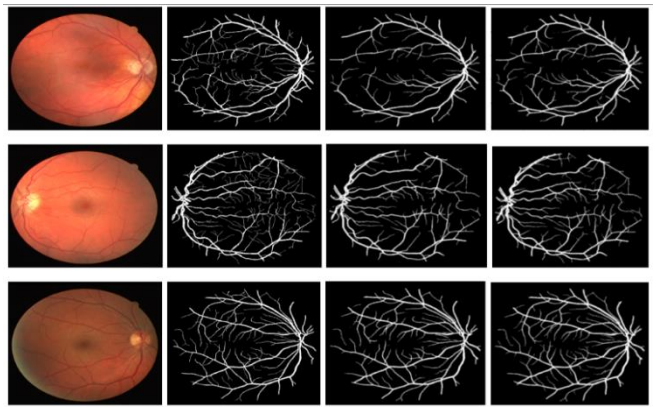


Fig. 3. Images 7, 11, and 15 from the DRIVE dataset for a qualitative performance comparison of ColonSegNet [11] and RetSegNet.

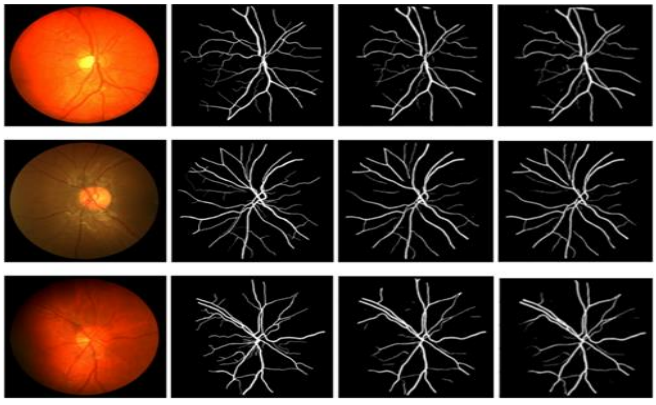


Fig. 4. Image numbers 2, 5, and 8 from the CHASE DB1 dataset are used for qualitative analysis of ColonSegNet [11] and RetSegNet.

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